Advanced Trigonometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Trigonometric identities: Pythagorean and reciprocal**

Given $\cos\theta = \frac{4}{5}$ and θ in the fourth quadrant, find $\sin\theta$ and $\tan\theta$.

$\sin^2\theta = 1 - \frac{16}{25} = \frac{9}{25}$. In the fourth quadrant, sine is negative: $\sin\theta = -\frac{3}{5}$. $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{-3/5}{4/5} = -\frac{3}{4}$.

2 **[Foundational]** • **Target: Sum and difference identities**

Use the sum identity to find the exact value of $\sin(75^\circ)$, using $\sin(75^\circ) = \sin(45^\circ + 30^\circ)$.

$\sin(45^\circ + 30^\circ) = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6}+\sqrt{2}}{4}$.

3 **[Foundational]** • **Target: Double angle identities**

Given $\sin\theta = \frac{3}{5}$ and θ in the first quadrant, find $\sin(2\theta)$ and $\cos(2\theta)$.

$\cos\theta = \frac{4}{5}$ (first quadrant, positive). $\sin(2\theta) = 2\sin\theta\cos\theta = 2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right) = \frac{24}{25}$.
$\cos(2\theta) = 1 - 2\sin^2\theta = 1 - 2\left(\frac{9}{25}\right) = 1 - \frac{18}{25} = \frac{7}{25}$.

4 **[Foundational]** • **Target: Graphs and transformations of trigonometric functions**

State the amplitude, period, and phase shift of $y = 2\sin\left(3x - \frac{\pi}{2}\right)$.

Amplitude: 2. Period: $\frac{2\pi}{3}$. Phase shift: solve $3x - \frac{\pi}{2} = 0$, giving $x = \frac{\pi}{6}$ (shift right by $\frac{\pi}{6}$).

5 **[Foundational]** • **Target: Applications: Law of Sines and Law of Cosines together**

Two ships leave a port at the same time, one traveling at 20° and the other at 80° (measured from north), each at 15 mph. Find the distance between the ships after 2 hours, to the nearest mile, using the Law of Cosines.

Each ship travels $15(2) = 30$ miles. The angle between their paths: $80^\circ - 20^\circ = 60^\circ$. Using the Law of Cosines: $d^2 = 30^2 + 30^2 - 2(30)(30)\cos 60^\circ = 900 + 900 - 1800(0.5) = 1800 - 900 = 900$. $d = \sqrt{900} = 30$ miles.

6 **[Foundational]** • **Target: Polar coordinates**

Convert the polar point $\left(4, \frac{\pi}{3}\right)$ to rectangular coordinates.

$x = r\cos\theta = 4\cos\left(\frac{\pi}{3}\right) = 4\left(\frac{1}{2}\right) = 2$. $y = r\sin\theta = 4\sin\left(\frac{\pi}{3}\right) = 4\left(\frac{\sqrt{3}}{2}\right) = 2\sqrt{3}$. Rectangular coordinates: $(2, 2\sqrt{3})$.

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7 [Foundational] • Target: A multi-part trigonometric identity verification problem

This question has two parts. (a) Use the identity $\tan\theta = \frac{\sin\theta}{\cos\theta}$ together with the double angle formulas to derive a formula for $\tan(2\theta)$ in terms of $\tan\theta$. (b) Use your formula to find $\tan(2\theta)$ given $\tan\theta = 3$.

(a) $\tan(2\theta) = \frac{\sin(2\theta)}{\cos(2\theta)} = \frac{2\sin\theta\cos\theta}{\cos^2\theta - \sin^2\theta}$. Dividing numerator and denominator by $\cos^2\theta$: $\tan(2\theta) = \frac{2\tan\theta}{1-\tan^2\theta}$. (b) $\tan(2\theta) = \frac{2(3)}{1-9} = \frac{6}{-8} = -\frac{3}{4}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Half-angle identities

Use the half-angle identity to find $\cos(15^\circ)$, given $\cos(30^\circ) = \frac{\sqrt{3}}{2}$.

Half-angle identity: $\cos\left(\frac{\theta}{2}\right) = \sqrt{\frac{1+\cos\theta}{2}}$. With $\theta = 30^\circ$: $\cos(15^\circ) = \sqrt{\frac{1+\frac{\sqrt{3}}{2}}{2}} = \sqrt{\frac{2+\sqrt{3}}{4}} = \frac{\sqrt{2+\sqrt{3}}}{2}$.

9 [Intermediate] • Target: Area of a triangle using trigonometry

Find the area of a triangle with sides 8 and 12, and included angle 50°, using $\text{Area} = \frac{1}{2}ab\sin C$.

Area $= \frac{1}{2}(8)(12)\sin(50^\circ) \approx \frac{1}{2}(96)(0.766) \approx 36.8$.

10 [Intermediate] • Target: Simplifying trigonometric expressions using identities

Simplify $\sin^2\theta + \cos^2\theta + \tan^2\theta$ in terms of $\sec\theta$.

Since $\sin^2\theta + \cos^2\theta = 1$, the expression becomes $1 + \tan^2\theta$. Using the Pythagorean identity $1 + \tan^2\theta = \sec^2\theta$, the simplified expression is $\sec^2\theta$.

11 [Intermediate] • Target: A multi-part navigation and triangulation problem

This question has three parts. A hiker walks 5 miles due east, then turns and walks 7 miles at a bearing of 40° north of east. (a) Find the angle between the two legs of the hike (measured as the interior angle at the turning point). (b) Use the Law of Cosines to find the straight-line distance from the start to the end point, to the nearest tenth of a mile. (c) Explain why this distance is less than the total distance walked (12 miles).

(a) The second leg deviates 40° from due east, so the interior angle at the turn (between reversing the first leg's direction and the second leg) is $180^\circ - 40^\circ = 140^\circ$. (b) $d^2 = 5^2 + 7^2 - 2(5)(7)\cos(140^\circ) = 25 + 49 - 70(-0.766) \approx 74 + 53.6 \approx 127.6$. $d \approx \sqrt{127.6} \approx 11.3$ miles. (c) The straight-line distance is always less than or equal to the sum of the individual leg lengths (a consequence of the triangle inequality) — walking in two different directions doesn't add distance as.

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12 [Intermediate] • Target: Verifying a trigonometric identity

Verify the identity $\frac{1-\cos^2\theta}{\sin\theta} = \sin\theta$.

Using the Pythagorean identity, $1-\cos^2\theta = \sin^2\theta$. Substituting: $\frac{\sin^2\theta}{\sin\theta} = \sin\theta$ (for $\sin\theta \neq 0$), confirming the identity holds.

13 [Intermediate] • Target: Simplifying trigonometric expressions using identities

Prove the trigonometric identity $\frac{1-\cos 2\theta}{\sin 2\theta} = \tan\theta$ for all values of θ where both sides are defined.

Step 1: Substitute the double-angle identities $\cos 2\theta = 1-2\sin^2\theta$ and $\sin 2\theta = 2\sin\theta\cos\theta$: $\frac{1-(1-2\sin^2\theta)}{2\sin\theta\cos\theta}$ Step 2: Simplify the numerator: $1-1+2\sin^2\theta = 2\sin^2\theta$ Step 3: Form the ratio and cancel common factor $2\sin\theta$ (for $\sin\theta eq 0$): $\frac{2\sin^2\theta}{2\sin\theta\cos\theta} = \frac{\sin\theta}{\cos\theta} = \tan\boldsymbol{\theta}$ ■

14 [Intermediate] • Target: A multi-part navigation and triangulation problem

Find all solutions to $2\cos^2\theta - \cos\theta - 1 = 0$ on the interval $[0, 2\pi)$.

Step 1: Factor the quadratic equation in $\cos\theta$: $(2\cos\theta+1)(\cos\theta-1)=0$ Step 2: Set each factor to zero: Case 1: $2\cos\theta+1=0 \Rightarrow \cos\theta = -\frac{1}{2}$. Cosine is negative in Quadrants II and III with reference angle $\frac{\pi}{3}$: $\theta = \pi - \frac{\pi}{3} = \boldsymbol{\frac{2\pi}{3}}$, $\theta = \pi + \frac{\pi}{3} = \boldsymbol{\frac{4\pi}{3}}$ Case 2: $\cos\theta - 1 = 0 \Rightarrow \cos\theta = 1$. On $[0, 2\pi)$, $\cos\theta = 1$ at $\theta = \boldsymbol{0}$. Final.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Verifying a trigonometric identity • Cross-Lesson: 5.01 Pre-Calculus Concepts

In ΔABC, $a = 6$, $b = 10$, and angle $A = 30^\circ$. Determine how many possible triangles exist, and find the possible value(s) of angle B in degrees.

Step 1 (Check SSA Ambiguity): Compute the altitude from vertex C: $h = b\sin A = 10\sin(30^\circ) = 10\left(\frac{1}{2}\right) = 5$ Compare the opposite side $a = 6$ with h and b: $htwodistincttrianglesexist.Step2(FindB_1andB_2xB_1 = \arcsin\left(\frac{5}{6}\right) \approx \boldsymbol{56.4}^\circ$ The obtuse supplementary angle is: $B_2 = 180^\circ - 56.4^\circ = \boldsymbol{123.6}^\circ$ Check validity: \(A + B_2 = 30^\circ + 123.6^\circ = 153.6^\circ

16 [Challenging] • Target: Trigonometric identities: Pythagorean and reciprocal • Cross-Lesson: 6.01 Advanced Coordinate Geometry

The sum of an arithmetic sequence is 150, with first term $a_1 = 5$ and last term $a_n = 25$. How many terms are in this sequence, and what is the common difference d?

Step 1 (Find n): Use the arithmetic series sum formula: $S_n = \frac{n}{2}(a_1 + a_n) \Rightarrow 150 = \frac{n}{2}(5+25) = 15n \Rightarrow n = \frac{150}{15} = \boldsymbol{10}$ Step 2 (Find d): Use the n-th term formula with $n = 10$: $a_{10} = a_1 + 9d \Rightarrow 25 = 5 + 9d \Rightarrow 9d = 20 \Rightarrow d = \boldsymbol{\frac{20}{9}}$

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17 [Challenging] • Target: Sum and difference identities • Cross-Lesson: 7.01 Advanced Plane Geometry

Calculate the exact value of $\cos(75^\circ)$ using angle addition, and evaluate $\tan(2\theta)$ given that $\tan\theta = \frac{2}{3}$.

Part 1: Exact value of $\cos(75^\circ)$: Express $75^\circ = 45^\circ + 30^\circ$ and apply the cosine sum identity: $\cos(45^\circ + 30^\circ) = \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ = \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right) = \frac{\sqrt{6}-\sqrt{2}}{4} = \frac{\sqrt{\mathbf{6}}-\sqrt{\mathbf{2}}}{\mathbf{4}}$. Part 2: Evaluate $\tan(2\theta)$: Apply the double-angle identity for tangent.

18 [Challenging] • Target: Double angle identities • Cross-Lesson: 8.01 Statistics, Counting & Probability

In triangle ABC, $a = 10$, $b = 16$, and $A = 30^\circ$. Determine whether $0, 1,$ or 2 distinct triangles can be formed, and calculate all possible values for angle B to the nearest tenth of a degree.

Step 1: Set up the Law of Sines for angle B: $\frac{\sin B}{b} = \frac{\sin A}{a} \Rightarrow \sin B = \frac{b \sin A}{a} = \frac{16 \sin 30^\circ}{10} = \frac{16(0.5)}{10} = \frac{8}{10} = 0.8$. Step 2: Find the acute angle solution B_1: $B_1 = \sin^{-1}(0.8) \approx \mathbf{53.1}^\circ$. Step 3: Test the obtuse supplementary angle B_2: $B_2 = 180^\circ - 53.1^\circ = 126.9^\circ$. Check triangle sum with given $A = 30^\circ$: \[A + B_2 = 30^\circ + 126.9^\circ = 156.9^\circ Conclusion: Exactly **2** d.

19 [Challenging] • Target: Graphs and transformations of trigonometric functions • Cross-Lesson: 1.01 Advanced Number Theory

In triangle ABC, $a = 7$, $b = 5$, and $\cos C = \frac{1}{5}$. (a) Find the exact length of side c. (b) Calculate the exact value of $\sin^2 C$ and the area of triangle ABC.

Part (a): Find side c using Law of Cosines: $c^2 = a^2 + b^2 - 2ab\cos C$ $c^2 = 7^2 + 5^2 - 2(7)(5)\left(\frac{1}{5}\right) = 49 + 25 - 14 = 60 \Rightarrow \mathbf{c = \sqrt{60} = 2\sqrt{15}}$. Part (b): Find $\sin^2 C$ and Area: 1. Pythagorean identity: $\sin^2 C = 1 - \cos^2 C = 1 - \left(\frac{1}{5}\right)^2 = 1 - \frac{1}{25} = \mathbf{\frac{24}{25}}$. 2. Since C is an angle of a triangle, $\sin C > 0 \Rightarrow \sin C = \frac{\sqrt{24}}{5} = \frac{2\sqrt{6}}{5}$. 3. Area formula.

20 [Challenging] • Target: Inverse trigonometric functions • Cross-Lesson: 2.01 Advanced Algebra & Systems

In triangle ABC, $a = 8$, $\angle B = 35^\circ$, and $\angle C = 75^\circ$. Use the Law of Sines to calculate the length of side b to two decimal places.

Step 1: Find angle A: $\angle A = 180^\circ - 35^\circ - 75^\circ = 70^\circ$. Step 2: Apply Law of Sines: $\frac{b}{\sin B} = \frac{a}{\sin A} \Rightarrow b = \frac{a \sin B}{\sin A} = \frac{8 \sin(35^\circ)}{\sin(70^\circ)}$. Step 3: Evaluate: $b = \frac{8(0.5736)}{0.9397} = \frac{4.5886}{0.9397} \approx \mathbf{4.88}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Trigonometric identities: Pythagorean and reciprocal

Given $\sin\theta = \frac{15}{17}$ and θ in the second quadrant, find $\cos\theta$.

a. $-\frac{15}{17}$

b. $\frac{15}{17}$

c. $-\frac{8}{17}$ ✓

d. $\frac{8}{17}$

$\sin^2\theta + \cos^2\theta = 1$, so $\cos\theta^2 = 1 - \sin\theta^2 = 1 - 225/289 = 64/289$. In the second quadrant, cosine is negative: $\cos\theta = -8/17$.

22 [Foundational] • Target: Sum and difference identities

Given $\cos\theta = -\frac{12}{13}$ and θ in the third quadrant, find $\sin\theta$.

a. $-\frac{12}{13}$

b. $\frac{12}{13}$

c. $-\frac{5}{13}$ ✓

d. $\frac{5}{13}$

$\sin^2\theta + \cos^2\theta = 1$, so $\sin\theta^2 = 1 - \cos\theta^2 = 1 - 144/169 = 25/169$. In the third quadrant, sine is negative: $\sin\theta = -5/13$.

23 [Foundational] • Target: Double angle identities

Two sides of a triangle measure 7 and 8, with an included angle of 120°. Find the length of the third side, using the Law of Cosines.

a. 14

b. 13 ✓

c. 12

d. 11

Using the Law of Cosines with $\cos(120^\circ) = -\frac{1}{2}$: $c^2 = 7^2 + 8^2 - 2(7)(8)\left(-\frac{1}{2}\right) = 49 + 64 + 56 = 169$, so $c = \sqrt{169} = 13$.

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24 [Foundational] • Target: Graphs and transformations of trigonometric functions

In a triangle, sides $a = 15$ and $b = 14$ meet at an included angle of 123°. Using the Law of Cosines, which expression gives side c?

a. $\sqrt{15^2 + 14^2 + 2(15)(14)\cos 123^\circ}$

b. $\sqrt{15^2 - 14^2 - 2(15)(14)\cos 123^\circ}$

c. $\sqrt{15^2 + 14^2 - 2(15)(14)\cos 123^\circ}$ ✓

d. $15^2 + 14^2 - 2(15)(14)\cos 123^\circ$

Law of Cosines: $c^2 = a^2 + b^2 - 2ab\cos C \Rightarrow c = \sqrt{15^2 + 14^2 - 2(15)(14)\cos 123^\circ}$.

25 [Foundational] • Target: Inverse trigonometric functions

Evaluate $\sin(45^\circ + 30^\circ)$ using a sum/difference formula. (variant 10)

a. $6 - 24$

b. 22

c. $2 - 64$

d. $6 + 24$ ✓

$(45^+ 30^) = 45^3 0^+ 45^3 0^= 6 + 24$.

26 [Foundational] • Target: Trigonometric equations with multiple solutions

Evaluate $\cos(60^\circ + 45^\circ)$ using a sum/difference formula.

a. $2 - 64$ ✓

b. $\frac{1}{2}$

c. 22

d. $6 - 24$

$(60^+ 45^) = 60^4 5^- 60^4 5^= 2 - 64$.

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27 [Foundational] • Target: Applications: Law of Sines and Law of Cosines together

In ΔABC, the side lengths opposite to angles A, B, and C are a, b, and c, respectively. If $\angle A - \angle B = 90^\circ$ and $a + b = \sqrt{2}c$, what is the measure of $\angle C$?

a. 30°

b. 45°

c. 60° ✓

d. 75°

We synthesize the Law of Sines with trigonometric sum-to-product and double-angle identities. 1. By the Law of Sines, $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$, which allows us to rewrite the given side relation in terms of angles: $\frac{a+b}{c} = \frac{\sin A + \sin B}{\sin C} = \sqrt{2}$ 2. Applying the sum-to-product identity to the numerator: $\sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$ Since the angles of a triangle sum to 180°.

28 [Foundational] • Target: Polar coordinates

In trapezoid $ABCD$, base $\overline{AB}$ is parallel to base $\overline{CD}$, and $\angle DAB = \angle ADC = 90^\circ$. The length of $\overline{AB}$ is x, the length of $\overline{CD}$ is $2x + 1$, and the length of the non-parallel side $\overline{BC}$ is $2x$. If the height $AD = 8$, what is the area of trapezoid $ABCD$?

a. 34

b. **64** ✓

c. 80

d. 128

To find the area of trapezoid $ABCD$, we first determine the value of the parameter x. 1. **Deconstruct the trapezoid:** Draw an altitude from vertex B perpendicular to $\overline{CD}$, intersecting $\overline{CD}$ at point E. Because $\angle DAB = \angle ADC = 90^\circ$ and $\overline{BE} \perp \overline{CD}$, quadrilateral $ABED$ is a rectangle. Therefore: $BE = AD = 8 \quad \text{and} \quad DE = AB = x$ 2. **Express the sides of right triangle ΔBEC:** Point E lies on $\overline{CD}$, so the re.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: A multi-part trigonometric identity verification problem

In ΔABC, the length of side AB is x centimeters, the length of side BC is $(x+3)$ centimeters, and the length of side AC is 7 centimeters. If the measure of $\angle B$ is 60°, what is the value of x?

a. 4

b. 5 ✓

c. 7

d. 8

To find the unknown parameter x, apply the Law of Cosines to ΔABC using the angle $\angle B = 60^\circ$ and its opposite side AC: $AC^2 = AB^2 + BC^2 - 2(AB)(BC)\cos(B)$ Substitute the given values into the formula: $7^2 = x^2 + (x+3)^2 - 2x(x+3)\cos(60^\circ)$ Since $\cos(60^\circ) = \frac{1}{2}$: $49 = x^2 + (x^2+6x+9) - 2x(x+3)\left(\frac{1}{2}\right)$ $49 = 2x^2 + 6x + 9 - (x^2 + 3x)$ $49 = x^2 + 3x + 9$ Subtract 49 from both sides to form a standard quadratic equation.

30 [Intermediate] • Target: Half-angle identities

In the standard (x, y) coordinate plane, the terminal ray of an angle θ in standard position passes through the point $(-3, 4)$. What is the value of $\sec\theta$?

a. $-\frac{5}{3}$ ✓

b. $-\frac{3}{5}$

c. $\frac{5}{4}$

d. $-\frac{4}{3}$

For an angle θ in standard position whose terminal ray passes through the point (x, y), the distance r from the origin is given by: $r = \sqrt{x^2 + y^2}$ Given $x = -3$ and $y = 4$: $r = \sqrt{(-3)^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$ By definition, $\cos\theta = \frac{x}{r}$, so the secant function is: $\sec\theta = \frac{1}{\cos\theta} = \frac{r}{x} = \frac{5}{-3} = -\frac{5}{3}$ Common errors: - Finding $\cos\theta = -\frac{3}{5}$ without taking the reciprocal leads to choice B. - Finding.

31 [Intermediate] • Target: Solving trigonometric equations that factor

In the figure below, ΔABC is a right triangle with right angle at B, and altitude $\overline{BD}$ is drawn perpendicular to hypotenuse $\overline{AC}$. If $AD = 4$ and $DC = 9$, what is the length of $\overline{BD}$?

a. $\sqrt{13}$

b. **6** ✓

c. 6.5

d. 13

By the Geometric Mean Theorem (Altitude Rule) for right triangles, the altitude to the hypotenuse is the geometric mean of the two segments of the hypotenuse: $BD^2 = AD \cdot DC$ Given $AD = 4$ and $DC = 9$: $BD^2 = 4 \cdot 9 = 36 \Rightarrow BD = \sqrt{36} = 6$

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32 [Intermediate] • Target: Area of a triangle using trigonometry

Two similar triangles have corresponding side lengths in ratio $4:6$. If the smaller triangle has area 8, what is the area of the larger triangle?

a. 10

b. 22

c. 14

d. **18** ✓

Area ratio is the square of the side ratio: $\left(\frac{6}{4}\right)^2 = \frac{36}{16}$. Larger area $= 8 \times \frac{36}{16} = 18$.

33 [Intermediate] • Target: Simplifying trigonometric expressions using identities

What are the equations of the asymptotes of the hyperbola $\frac{(x-1)^2}{9} - \frac{(y+2)^2}{4} = 1$?

a. $y + 2 = \pm\frac{2}{3}(x-1)$ ✓

b. $y + 2 = \pm\frac{3}{2}(x-1)$

c. $y = \pm\frac{2}{3}x$

d. $y - 2 = \pm\frac{2}{3}(x+1)$

Center $(h, k) = (1, -2)$, slopes $\pm\frac{b}{a} = \pm\frac{2}{3}$. Asymptotes: $y + 2 = \pm\frac{2}{3}(x-1)$.

34 [Intermediate] • Target: A multi-part navigation and triangulation problem

An ellipse has equation $\frac{x^2}{36} + \frac{y^2}{20} = 1$. What are the coordinates of its foci?

a. $(\pm 5, 0)$

b. $(\pm 4, 0)$ ✓

c. $(0, \pm 4)$

d. $(\pm 6, 0)$

$c^2 = a^2 - b^2 = 36 - 20 = 16 \Rightarrow c = 4$. Since $a^2 > b^2$, foci lie along the x-axis at $(\pm 4, 0)$.

35 [Intermediate] • Target: Verifying a trigonometric identity

In ΔABC, $\angle A = 60^\circ$, $\angle B = 80^\circ$, and side $a = 5$. Find side b (to one decimal place) using the Law of Sines.

a. **5.7** ✓

b. 6.7

c. 7.7

d. 4.7

By the Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B}$, so $b = \frac{5\sin 80^\circ}{\sin 60^\circ} \approx 5.7$.

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36 [Intermediate] • Target: Trigonometric identities: Pythagorean and reciprocal

Find the inverse of the matrix $M = (5, 2; 2, 1)$.

a. $(-1, 2; 2, -5)$

b. $(1, 2; 2, 5)$

c. $(5, -2; -2, 1)$

d. $(1, -2; -2, 5)$ ✓

$\det(M) = (5)(1) - (2)(2) = 1$. Inverse $M^{-1} = (d, -b; -c, a) = (1, -2; -2, 5)$.

37 [Intermediate] • Target: Sum and difference identities

Evaluate $(1 + i)^4$ using De Moivre's Theorem.

a. 4

b. **-4** ✓

c. 16

d. -4i

$1 + i = \sqrt{2}\,\text{cis}(\pi/4)$. Then $(1 + i)^4 = (\sqrt{2})^4 \text{cis}(1\pi) = 2^2(-1) = -4$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Double angle identities • Cross-Lesson: 8.01 Statistics, Counting & Probability

Evaluate the exact trigonometric value: $\sin\left(\frac{\pi}{6}\right)$.

a. $\frac{\sqrt{3}}{2}$

b. $\frac{\sqrt{2}}{2}$

c. 1

d. $\frac{1}{2}$ ✓

Using reference angles on the unit circle, $\frac{\pi}{6} = 30^\circ$. The exact value is $\frac{1}{2}$.

39 [Challenging] • Target: Graphs and transformations of trigonometric functions • Cross-Lesson: 1.01 Advanced Number Theory

Find the area of a triangle with base 12 and height 16.

a. 192

b. **96** ✓

c. 108

d. 80

Area $= \frac{1}{2}bh = \frac{1}{2}(12)(16) = 96$.

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40 [Challenging] • Target: Inverse trigonometric functions • Cross-Lesson: 2.01 Advanced Algebra & Systems

Prove the identity $\sin 3A = 3\sin A - 4\sin^3 A$ using the angle addition formula.

a. Not an identity

b. Only for $A < \pi/2$

c. Use calculus

d. $\sin(2A + A) = \sin 2A\cos A + \cos 2A\sin A = 2\sin A\cos^2 A + (1 - 2\sin^2 A)\sin A = 3\sin A - 4\sin^3 A$. ✓ ✓

Applying Trigonometric Functions principles, Standard derivation. Therefore, the verified correct option is $\sin(2A + A) = \sin 2A\cos A + \cos 2A\sin A = 2\sin A\cos^2 A + (1 - 2\sin^2 A)\sin A = 3\sin A - 4\sin^3 A$. ✓.

41 [Challenging] • Target: Trigonometric equations with multiple solutions • Cross-Lesson: 3.01 Advanced Functions

In ΔABC, $a = 8$, $b = 10$, $c = 12$. Find $\angle A$ (to one decimal place) using the Law of Cosines.

a. **41.4°** ✓

b. 46.4°

c. 138.6°

d. 36.4°

By the Law of Cosines: $\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{100 + 144 - 64}{240} \approx 0.75$, so $A = \cos^{-1}(0.75) \approx 41.4°$.

42 [Challenging] • Target: Applications: Law of Sines and Law of Cosines together • Cross-Lesson: 5.01 Pre-Calculus Concepts

In ΔABC, $a = 5$, $b = 6$, $c = 7$. Find $\angle A$ (to one decimal place) using the Law of Cosines. (trial 2)

a. 49.4°

b. 135.6°

c. **44.4°** ✓

d. 39.4°

By the Law of Cosines: $\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{36 + 49 - 25}{84} \approx 0.714$, so $A = \cos^{-1}(0.714) \approx 44.4°$.

43 [Challenging] • Target: Polar coordinates • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Using a double-angle identity, evaluate $\cos^2\theta - \sin^2\theta$ when $\theta = 15°$.

a. -1

b. $-\frac{\sqrt{3}}{2}$

c. $\frac{\sqrt{2}}{2}$

d. $\frac{\sqrt{3}}{2}$ ✓

$\cos^2\theta - \sin^2\theta = \cos(2\theta)$; at $\theta = 15°$, $2\theta = 30°$, so the value is $\frac{\sqrt{3}}{2}$.

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44 [Challenging] • Target: A multi-part trigonometric identity verification problem • Cross-Lesson: 7.01 Advanced Plane Geometry

If $\cos\theta = \frac{1}{2}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{1}{4}$

b. $\frac{1}{2}$ ✓

c. $\frac{3}{5}$

d. $-\frac{1}{2}$

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-\left(\frac{1}{2}\right)}{2}} = \frac{1}{2}$.

45 [Challenging] • Target: Half-angle identities • Cross-Lesson: 8.01 Statistics, Counting & Probability

In ΔABC, $b = 10$, $c = 7$, and $\angle A = 120^\circ$. Find side a (to one decimal place) using the Law of Cosines.

a. 13.8

b. **14.8** ✓

c. 15.8

d. 12.2

By the Law of Cosines: $a^2 = b^2 + c^2 - 2bc\cos A = 10^2 + 7^2 - 2(10)(7)\cos 120^\circ \approx 219.0$, so $a \approx 14.8$.

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